

Ghost Kitchen Operational Optimization

A Structured Case Study on Logistics, Menu Engineering, and Performance Analytics

1. Project Overview

This project evaluates the performance of a high-volume virtual restaurant (ghost kitchen) brand operating across urban fulfillment corridors. In a ghost kitchen model, businesses lack public dining storefronts and interact with customers entirely through digital applications and third-party delivery services. Consequently, operational profitability is tied to internal kitchen preparation times, courier speeds, and customer feedback management. The goal of this case study is to build an analytical data pipeline that moves raw transactional data into a single-screen executive workspace. By isolating hidden delivery latencies, identifying underperforming items, and analyzing peak demand hours, this analysis helps stakeholders improve logistical efficiency and capture higher profit margins.

2. Core Analytical Tools

The project leverages a specialized, two-tier data stack to clean data and generate executive insights:

- SQL Database Engine:** Used to handle core data management tasks. SQL queries aggregate relational records, perform schema connections across multiple business tables, calculate date-time differences, and extract filtered operational metrics.
- LibreOffice Calc (Spreadsheet):** Serves as the primary analytics workspace and visualization front-end. It houses the structured data tabs, performs statistical evaluations, removes data inconsistencies, and displays flat 2D executive dashboards for non-technical leadership.

3. Dataset Summary

The database architecture contains the underlying transactional information, organized into four relational tables that track key aspects of the business:

A. Core Data Tables

Table Name	Data Field Properties	Business Analytical Scope
Menu_Items	item_id, item_name, category, price, cost_to_make	Tracks item pricing against cost structures to calculate profit margins.
Delivery_Partners	driver_id, driver_name, vehicle_type	Identifies courier personnel and their vehicle classifications.
Orders	order_id, customer_id, item_id, driver_id, timestamps, status	The main log tracking timestamps from order creation to delivery.
Customer_Reviews	review_id, order_id, rating, review_text	Logs customer sentiment data based on a standard 1-to-5 star scale.

B. Baseline Operational Performance Metrics

Aggregating these datasets yields the following verified baseline metrics, which establish the operational benchmarks for the business model:

Key Metric Property	Consolidated Value	Contextual Meaning
Orders Completed	972 Orders	Successful revenue-generating fulfillment events.

Key Metric Property	Consolidated Value	Contextual Meaning
Orders Cancelled	28 Orders	Fulfillment breakdowns causing direct ingredient and revenue loss.
Orders Reviewed	324 Reviews	The feedback sample used to track customer sentiment.
Average Customer Rating	3.0 / 5.0 Stars	A plateaued satisfaction score indicating delivery friction.
Total Project Revenue	Rs. 13,145.38	Gross top-line intake generated by completed menu orders.
Total Project Net Profit	Rs. 8,839.38	Net margin capture after deducting raw ingredient preparation costs.

4. Analysis & Findings

A. Logistics Bottleneck Identification

To understand what holds back the 3.0-star customer satisfaction score, order lifecycles were split into internal and external phases. The analysis shows that **Average Transit Time scales significantly higher than Kitchen Preparation Time**. This demonstrates that the main operational challenge lies in courier delivery dispatch and road transit, rather than internal kitchen bottlenecks.

B. Delivery Fleet Performance Variations

Cross-referencing transit times with courier vehicle types reveals clear performance trends: Two-Wheeled Vehicles (Scooters and E-Bikes) display short and optimized travel durations within urban corridors, effectively bypassing street congestion. Conversely, Four-Wheeled Vehicles (Cars) introduce major transit delays due to traffic gridlock and parking constraints in high-density delivery zones, making them less efficient for fast turnarounds.

C. Menu Engineering Strategy

While the overall profit of Rs. 8,839.38 indicates a healthy margin structure, individual items show distinct volume and profitability trends. Popular items like the *Premium Truffle Burger* combine high transaction volumes with strong profit percentages. Conversely, lower-performing items require menu re-pricing or ingredient cost changes to protect overall profit margins.

D. Customer Sentiment Gaps

Comparing total completed orders against review submissions reveals an engagement gap: **only 33.3% of orders receive customer feedback** (324 reviews out of 972 orders). This lack of participation leaves a notable blind spot for tracking quality issues like food temperature and packaging condition at the time of delivery.

5. Strategic Actions & Recommendations

1. Enforce a Two-Wheeled Fleet Dispatch Priority

Restrict the use of cars on short-to-medium urban delivery routes. Prioritizing scooters and e-bikes allows couriers to navigate city traffic more effectively, helping reduce average transit times and ensure food arrives hot.

2. Launch Customer Feedback Prompts

Introduce small post-delivery rewards or clear in-app prompts to encourage review submissions. Increasing feedback beyond the current 33.3% baseline provides the specific insights needed to resolve problems behind the 3.0-star satisfaction score.

3. Align Staffing with Peak Wave Surges

Structure driver shifts and kitchen preparation schedules to match lunch and dinner demand waves. Increasing driver density right before these known peaks helps smooth out delivery spikes and lowers order cancellation risks.